

Temperature influences on California rainbow trout physiological performance

C. A. Myrick* and J. J. Cech, Jr.

Department of Wildlife, Fish, and Conservation Biology, University of California Davis, 1 Shields Avenue, Davis, California, 95616-8751, USA; *Present address: Department of Fishery and Wildlife Biology, Colorado State University, Fort Collins, CO 80523, USA (Phone: (530)752-8659; Fax: (530)752-4154; E-mail: camyrick@ucdavis.edu.)

Accepted: December 10, 1999

Key words: swimming velocity, thermal tolerance, *Oncorhynchus mykiss*, metabolism, food consumption, growth

Abstract

Although thermal influences on the physiology of rainbow trout (*Oncorhynchus mykiss*) have been widely studied, there is little information about the responses of different genetic strains to temperatures. The effects of water temperature (10, 14, 19, 22, and 25 °C) on food consumption rate, growth rate, gross conversion efficiency, resting routine oxygen consumption rate, upper critical thermal tolerance and critical swimming velocity were investigated in juvenile rainbow trout of the Eagle Lake (*O. m. aquilarum*) subspecies and the Mt. Shasta strain. No strain-related differences in conversion efficiency, oxygen consumption rates, thermal tolerance or swimming performance were observed in 1995 (19, 22, and 25 °C) or 1996 (10, 14, and 19 °C), but Mt. Shasta strain trout grew faster at highest temperatures than did Eagle Lake trout. Food consumption rates, growth rates, conversion efficiency, and oxygen consumption rates declined at the extreme temperatures (10 and 25 °C) in both Eagle Lake and Mt. Shasta trout. Swimming performance was temperature-independent between 10 and 19 °C (overall mean: 5.43 body lengths s⁻¹).

Introduction

Temperature influences physiological variables in fishes including food consumption, growth, and metabolic rates (Brett and Groves 1979). Much research into temperature effects has been conducted on salmonid fishes, especially rainbow trout (*Oncorhynchus mykiss*) (Brett and Groves 1979; Wood and McDonald 1997) though little attention has been directed at comparisons of temperature-related effects between strains. Strain-specific differences in physiological performance have been reported for a number of salmonids, including rainbow trout (Thomas and Donahoo 1977; Hawkins and Quinn 1996), sockeye salmon (*O. nerka*) (Taylor and Foote 1991), and coho salmon (*O. kisutch*) (Taylor and McPhail 1985).

Individual-based bioenergetic population models (Ney 1993) may accurately predict the responses of rainbow trout over a relatively wide range (Hanson

et al. 1997), but the physiological effects of chronically elevated temperatures have received little attention, so the models may not accurately predict responses at temperatures exceeding 20 °C. In California water temperatures above the optimal range, approaching the reported incipient lethal limit for rainbow trout, are commonplace. This, together with possible inter-strain or inter-population differences in physiological responses to temperature warrants the investigation of responses of fish at temperatures close to their tolerance extremes.

California is home to 5–10 endemic rainbow trout subspecies (Moyle 1976; Nielsen 1994; McEwan and Jackson 1996), each of which has presumably evolved its particular life history for distinctive environmental conditions. California also hosts several distinct hatchery rainbow trout strains (Busack and Gall 1980). California's size and geologic diversity may impose

both elevational and latitudinal variation in population responses to temperature (Conover 1990).

Eagle Lake rainbow trout (*O. m. aquilarum*) are native to the alkaline Eagle Lake (1560 m elevation; Lassen County, CA) and historically spawned in Pine Creek, the major lake tributary. Intense water diversion, logging, grazing, and road and railroad construction activity led to the severe degradation of Pine Creek spawning habitat. Consequently, the California Department of Fish and Game (CDFG) commenced trapping the entire spawning run in 1956 and has since artificially spawned and reared both wild-caught and domestic Eagle Lake rainbow trout (Busack and Gall 1980). Eagle Lake rainbow trout are afforded subspecific status (Behnke 1992). Mt. Shasta rainbow trout are highly inbred generic hatchery rainbow trout derived from a cross between fish from Meader's Trout Farm (Pocatello, Idaho) and Hot Creek strain rainbow trout (Busack and Gall 1980).

We hypothesized that strain-related differences might exist in physiological responses to temperatures between Eagle Lake and Mt. Shasta rainbow trout. This was tested by comparing food consumption, growth, oxygen consumption, thermal tolerances, and swimming performance (1996 only) of the two strains at 19, 22, and 25 °C in 1995, and 10, 14, and 19 °C in 1996. The latter temperature range reflects temperatures encountered by rainbow trout in relatively undisturbed watersheds, whereas the former is representative of modified California watersheds (Moyle and Nichols 1974).

Materials and methods

Fish were first used in a 30-d food consumption and growth experiment, and were then tested for thermal tolerance, oxygen consumption, or swimming performance.

Fish collection and holding

Age-0 Eagle Lake and Mt. Shasta strain rainbow trout (1995 mean weights: 3.5 ± 0.05 g and 4.2 ± 0.06 g, respectively; 1996 mean weights: 2.4 ± 0.06 and 3.0 ± 0.08 g) were obtained from the Mt. Shasta State Fish Hatchery (1083 m above sea level; rearing temperature: 14 °C). Fish were transported to the University of California, Davis (UCD) and gradually acclimated (1 °C d^{-1}) to unchlorinated, air-equilibrated well water at 19, 22, and 25 °C in

1995, and 10, 14, and 19 °C in 1996. Water characteristics were: total dissolved solids 390 mg l^{-1} , total suspended solids $<5.0\text{ mg l}^{-1}$, total alkalinity 300 mg l^{-1} , ammonia-nitrogen $<0.1\text{ mg l}^{-1}$, pH 7.8 and hardness 320 mg l^{-1} . Dissolved oxygen levels in the flow-through experimental tanks, which also incorporated continuous aeration, were never below 90% air-saturation; weekly tests for total ammonia-N detected none (0.1 mg l^{-1} detection limit). Trout were stocked in 110-l round fiberglass tanks (5 replicates per temperature/strain treatment) at a density of 30 fish tank $^{-1}$ (1995) or 28 fish tank $^{-1}$ (1996). Water flow rates were set at 4 l min^{-1} in both years. In 1996, angled spray bars were used to adjust current velocity to 1 body length per second (BL s^{-1}), and the current direction was reversed every 5 days. Current velocity in 1995 was not uniform between tanks, but never exceeded 1 BL s^{-1} . Fish were held indoors in a facility that received both natural light (through translucent roof panels) and artificial lighting set to the natural photoperiod (latitude 38.55° N; July–September).

Food consumption

Fish were fed to apparent satiation twice daily on floating commercial trout pellets (Silvercup; manufacturer's analysis: crude protein 40%, crude fat 10%, crude fiber 3%, ash 12%, sodium 2%). Fish were initially fed the previous day's ration plus 0.2 g and a check was made after 5 min to see if all the food had been consumed. If all food was consumed, food was added in 0.5 g increments every 5 min until uneaten food remained. The amount of food consumed was quantified after each meal by collecting the uneaten pellets and consumption rate in percent body weight of food consumed per day was calculated for each tank (Wurtsbaugh and Davis 1977b) using:

$$\begin{aligned} \text{consumption rate (\% body weight d}^{-1}\text{)} &= \\ &= \frac{C}{0.5(W_1 + W_2)t} \times 100, \end{aligned} \quad (1)$$

where W_1 = initial estimated dry weight of a group of fish, W_2 = final dry weight of the group of fish, t = duration of the experiment in days (30 days) and, C = dry weight of food consumed. A subsample of 10 fish per treatment were over-anesthetized at the start and end of the experiment and dried for at least 5 d at 60 °C to determine mean initial, and final dry weights, respectively.

Short-term growth

Fish were fasted for 24 h before being anesthetized (50 mg l⁻¹ MS-222; 3 g l⁻¹ NaCl; 0.1 g l⁻¹ NaHCO₃), weighed to the nearest 0.1 g on a calibrated electronic balance and standard length (SL), fork length (FL), and total length (TL) measured to the nearest mm. Total lengths were used to set tank current velocities in 1996. In 1995, all fish were weighed and measured on day 0 and day 30 of the experiment; in 1996 we also weighed and measured the fish on day 10 and day 20. Initial (W_1) and final (W_2) dry weights (g) for each time interval (t) were used to calculate the relative growth rate (% body weight per day) for each tank (Wurtsbaugh and Davis 1977b) using:

$$\begin{aligned} \text{relative growth rate (\% body weight d}^{-1}\text{)} &= \\ &= \frac{W_2 - W_1}{0.5(W_1 + W_2)t} \times 100. \end{aligned} \quad (2)$$

Gross food conversion efficiencies (GCE, %) were calculated using:

$$\begin{aligned} \text{gross conversion efficiency (GCE, \%)} &= \\ &= \frac{W_2 - W_1}{\sum_{30}^1 C_i} \times 100, \end{aligned} \quad (3)$$

where W_2 and W_1 are as above, and C_i is the daily amount of food consumed.

Routine oxygen consumption

Resting routine oxygen consumption (Cech 1990) was measured in 900-ml cylindrical glass respirometers. Oxygen consumption experiments were conducted on 24-h fasted fish in (rearing) temperature-controlled water baths. Individual trout were placed in respirometers between 07:00 and 08:00 h with a 100 ml water min⁻¹ flushing rate, covered sufficiently to shield them from laboratory-induced stimulation, and allowed 4 h to reach a quiescent state (determined from a pilot study). An initial water sample was taken, the respirometer was sealed, and the water sample's partial pressure of oxygen (PO₂) was measured using a Radiometer PHM71/D616/E5046 thermostatted O₂ analyzer system. PO₂ values were converted to O₂ contents (mg O₂ l⁻¹) using a solubility nomogram. After 25 to 45 min, the final water sample was taken for analysis and the respirometer unsealed. Oxygen consumption rates for these 'closed' respirometers were calculated following Cech (1990). ANCOVA was used to compare between treatments' regression expressions of log oxygen consumption rate (mg h⁻¹) to weight.

Upper thermal tolerance

Acute upper thermal tolerance, expressed as critical thermal maxima, was evaluated on individual 24-h fasted trout as described by Young and Cech (1996). Water temperature was increased by 0.3 °C min⁻¹ and loss of equilibrium was used as the endpoint for tests.

Swimming performance

In 1996 swimming performance was assessed by measuring the critical swimming velocity (Brett 1964) of individual fish in a 152-l Brett-type swimming flume. Fatigue was taken as the point when fish would not move off the rear screen of the flume, even after reductions in current velocity and gentle prodding with a nylon rod. A short, 10-min interval and 1 BL s⁻¹ velocity increment were used. Rainbow trout used in pilot swimming trials in 1995 performed poorly (critical swimming velocity <1 BL s⁻¹ at 19 °C), and no swimming tests were conducted in 1995.

Data analyses

Data from each year were evaluated separately because of differences in rearing methods. Differences among treatment means were analyzed using Student's *T*-tests or ANOVA. Post-hoc comparisons were made using Student–Newman–Keuls method. The significance level for all statistics before data collection was set at $P < 0.05$.

Results

Eagle Lake and Mt. Shasta trout food consumption tended to increase with temperature over the 10 to 19 °C range. Increases were significant between 14 and 19 °C for the Eagle Lake trout and between 10 and 14 °C for the Mt. Shasta trout, despite confounding effects of size which tended to mask the temperature effect (Table 1). In contrast, Eagle Lake and Mt. Shasta trout consumed less food as the temperature increased from 19 to 25 °C (Table 1). Eagle Lake trout consumed significantly less food than Mt. Shasta trout at 19 °C in 1995; no other differences between strains were observed in either year (Table 1).

All treatments' final wet weights were significantly higher than their initial wet weights (Table 1). However, the increase in the 25 °C fishes' wet weight was due to an increase in water content (data not shown). Mean Eagle Lake and Mt. Shasta trout relative growth

Table 1. Temperature effects on ELRT and MSRT growth, food consumption, and conversion efficiency. All final weights are significantly larger than corresponding initial weights. Statistical differences between treatment means are indicated by * between strains at the same temperature and by different numbers for the same strains between temperatures

Year	Temperature (°C)	Treatment	n	Consumption rate (% body weight d ⁻¹)	Initial weight (g)	Final weight (g)	Growth Rate (% body weight d ⁻¹)	Gross conversion efficiency (%)	Survival (%)
1996	10	Eagle Lake	5	8.73 ± 0.67 ¹	2.3 ± 0.1 ^{1*}	5.2 ± 0.3 ^{1*}	2.56 ± 0.07 ¹	30.0 ± 2.31 ¹	98.6 ± 1.4 ¹
		Mt. Shasta	5	7.98 ± 0.16 ¹	4.1 ± 0.2 ^{1*}	9.3 ± 0.5 ^{1*}	2.56 ± 0.05 ¹	32.1 ± 0.80 ¹	99.3 ± 0.7 ¹
	14	Eagle Lake	5	9.41 ± 0.38 ¹	2.4 ± 0.1 ¹	7.0 ± 0.3 ²	3.22 ± 0.07 ²	34.4 ± 1.49 ¹	97.9 ± 0.9 ¹
		Mt. Shasta	5	9.68 ± 0.23 ²	2.7 ± 0.3 ²	8.3 ± 1.1 ¹	3.31 ± 0.11 ²	34.4 ± 1.82 ²	96.4 ± 2.8 ¹
	19	Eagle Lake	5	9.91 ± 0.19 ²	2.4 ± 0.1 ¹	7.5 ± 0.2 ²	3.32 ± 0.14 ²	33.4 ± 0.84 ¹	97.9 ± 1.4 ¹
		Mt. Shasta	5	10.80 ± 0.13 ¹	2.3 ± 0.0 ²	7.6 ± 0.3 ¹	3.56 ± 0.06 ²	32.9 ± 0.51 ^{1,2}	99.3 ± 0.7 ¹
1995	19	Eagle Lake	5	9.97 ± 0.61 ^{1*}	3.6 ± 0.1 ¹	15.3 ± 0.9 ^{1*}	3.62 ± 0.17 ¹	37.2 ± 4.13 ¹	78.7 ± 0.8 ¹
		Mt. Shasta	5	11.40 ± 0.22 ^{1*}	4.3 ± 0.0 ¹	20.4 ± 0.5 ^{1*}	3.74 ± 0.04 ¹	32.9 ± 0.99 ¹	75.3 ± 0.8 ¹
	22	Eagle Lake	5	8.66 ± 0.08 ²	3.6 ± 0.0 ¹	11.8 ± 0.4 ^{2*}	2.97 ± 0.12 ^{2*}	34.3 ± 1.57 ¹	79.3 ± 1.2 ¹
		Mt. Shasta	5	9.21 ± 0.16 ²	4.5 ± 0.3 ¹	16.7 ± 0.5 ^{2*}	3.51 ± 0.06 ^{1*}	38.1 ± 0.95 ¹	80.7 ± 0.7 ¹
	25	Eagle Lake	5	7.73 ± 0.39 ²	3.5 ± 0.0 ¹	4.4 ± 0.2 ³	-0.35 ± 0.21 ^{3*}	-4.5 ± 2.56 ²	72.0 ± 3.1 ¹
		Mt. Shasta	5	7.46 ± 0.42 ³	4.2 ± 0.1 ¹	5.8 ± 0.2 ³	0.05 ± 0.12 ^{2*}	0.8 ± 1.68 ²	74.0 ± 3.9 ¹

rates tended to increase as temperature increased from 10 to 19 °C (significant increase from 10 to 14 °C), and decreased to near zero as the temperature increased from 19 to 25 °C (Table 1). Despite a pre-experiment prophylactic treatment, the residual mortality from a *Costia* infection of both strains resulted in lower survival rates in 1995 than in 1996. Survival rates within years did not differ between treatments.

Eagle Lake and Mt. Shasta trout gross conversion efficiencies showed no clear relationship with temperature between 10 and 19 °C (Table 1). Similarly, no temperature-related differences were observed at 19 or 22 °C, but there was a significant decrease in gross conversion efficiency at 25 °C (Table 1). No differences between the two trout strains were detected.

The slopes of the regression lines of log oxygen consumption rates on log body weight were not significantly different (mean slope: 0.98; $P = 0.74$), so oxygen consumption rates were adjusted to body weight^{0.98} to minimize the influence of size differences (range: 5.0 to 24.5 g) among and within treatments (Cech 1990). There were no differences in mean oxygen consumption rate (mg O₂ h⁻¹) or mass-adjusted oxygen consumption rate (mg O₂ h⁻¹ g^{-0.98}) either within or between strains at 10, 14 or 19 °C (Table 2). Eagle Lake trout at 19 and 22 °C consumed similar amounts of oxygen (mg h⁻¹), and consumed more oxygen than Eagle Lake trout at 25 °C (Ta-

ble 2). Mt. Shasta trout consumed less oxygen at 25 °C than at 19 °C, whereas oxygen consumption rates at 22 °C were intermediate. No differences were detected between the two strains, and the mass-adjusted oxygen consumption rates were unaffected by strain and temperature (Table 2).

Both strains tended to show higher thermal tolerances with higher rearing temperatures in both years (Table 3). There were no strain-related differences in critical thermal tolerance for fish acclimated to the same temperature (Table 3).

There were no significant differences between the strains' length-specific critical swimming velocity the only differences in critical swimming velocity (m s⁻¹) being due to the smaller size (shorter by 18%) of the 10°C Eagle Lake trout (Table 4).

Discussion

Our study of the effects of temperature on Eagle Lake and Mt. Shasta rainbow trout physiology revealed few strain-related differences, except that Mt. Shasta trout grew faster than Eagle Lake trout at the warmest temperatures (22 and 25 °C). The temperature effects on food consumption, growth, and thermal tolerance are in accord with literature reports recently reviewed by Jobling (1997). Both strains ate less food and grew slowest at 10 °C. The increasing trend in both strains' food consumption and growth rates from 14 to 19 °C

Table 2. Temperature effects on Eagle Lake and Mt. Shasta rainbow trout mean ($\pm$ SEM) routine oxygen consumption rates (MO_2). Statistical differences between treatment means are indicated by * between strains at the same temperature and by different numbers for the same strains between temperatures

Year	Temperature (°C)	Strain	Sample size	Weight (g)	MO_2 (mg h^{-1})	MO_2 ($\text{mg/h/g}^{-0.98}$)
1996	10	Eagle Lake	18	$6.9 \pm 0.51^{1*}$	1.85 ± 0.22^1	0.28 ± 0.03^1
		Mt. Shasta	19	$10.0 \pm 0.76^{1*}$	1.96 ± 0.21^1	0.22 ± 0.02^1
	14	Eagle Lake	20	7.2 ± 0.59^1	1.73 ± 0.18^1	0.30 ± 0.02^1
		Mt. Shasta	19	7.5 ± 0.71^1	1.70 ± 0.17^1	0.25 ± 0.03^1
	19	Eagle Lake	20	7.5 ± 0.80^1	1.95 ± 0.26^1	0.26 ± 0.02^1
		Mt. Shasta	20	8.2 ± 0.75^1	1.88 ± 0.24^1	0.24 ± 0.02^1
1995	19	Eagle Lake	18	14.1 ± 1.49^1	4.25 ± 0.64^1	0.30 ± 0.02^1
		Mt. Shasta	19	24.5 ± 4.86^1	6.30 ± 1.59^1	0.27 ± 0.03^1
	22	Eagle Lake	19	$13.4 \pm 1.99^{1,2}$	4.15 ± 0.58^1	0.33 ± 0.02^1
		Mt. Shasta	19	$15.0 \pm 1.59^{1*}$	$3.90 \pm 0.44^{1,2}$	0.28 ± 0.01^1
	25	Eagle Lake	9	5.0 ± 0.51^2	1.50 ± 0.09^2	0.32 ± 0.02^1
		Mt. Shasta	5	5.4 ± 0.67^2	1.52 ± 0.20^2	0.30 ± 0.04^1

suggests that the optimal temperature for growth lies between these 2 temperatures, which brackets reported optimal temperatures for rainbow trout (Hokanson et al. 1977; Briggs and Post 1997). As temperature increased from 19 to 25 °C, both strains consumed less food and grew less, as expected, with the greatest reduction at 25 °C, which approaches the reported upper incipient lethal temperatures (25 to 26 °C) for rainbow trout (Cherry et al. 1977; Hokanson et al. 1977). Interestingly, oxygen consumption rates were not closely coupled to food consumption or growth rates increases with increasing temperature (Cech et al. 1990; Jobling 1994). Our results probably reflect a shift in energy allocation with increasing temperature, with metabolism receiving a greater portion of the ingested energy than at lower temperatures.

Wurtsbaugh and Davis (1977b) reported that 1 to 1.2 g steelhead (anadromous rainbow trout) reared for 25 d at 16.2, 19.5, and 22.5 °C had consumption rates of 14.3, 15.7, and 13.4% body weight d^{-1} . These values are $\approx 5\%$ higher than those reported in this study and may result from the disparity in size or diet quality (Brett et al. 1969; Wurtsbaugh and Davis 1977a). Growth rates for our trout closely matched those reported by Alsop and Wood (1997) but were lower than the growth rates measured by Hokanson et al. (1977) in rainbow trout from Lake Superior (3.66, 4.36, 4.83, and 3.94% body weight d^{-1} in 0.2–0.3 g trout at temperatures of 10, 15, 19, and 22 °C, respectively). These slightly higher values may have resulted from the size

difference between the Lake Superior fish and those used in this study. Wurtsbaugh and Davis (1977b) measured growth rates of 2.6, 2.4, and 1.7% body weight d^{-1} in juvenile steelhead reared at 16.2, 19.5 and 22.5 °C, respectively. Consistent with results reported by Hokanson et al. (1977), Reid et al. (1998), and Dockray et al. (1996, 1998), our data demonstrate that rainbow trout are capable of growing well at temperatures near 22 °C but that growth rates declined rapidly as temperatures approach 25 °C. Both Eagle Lake and Mt. Shasta trout were, however, able to maintain weight at 25 °C for 30 d. This trait may allow them to survive short (<1 month) periods of sublethal high temperatures in California streams.

Eagle Lake and Mt. Shasta trout food consumption, growth, and metabolism followed similar trends, but the Mt. Shasta trout's values tended to be higher. These higher values (i.e., greater performance) may stem from a slight countergradient variation effect, because the source of the Mt. Shasta broodstock is at a slightly higher latitude than Eagle Lake's, the source of the Eagle Lake broodstock. However, the latitudinal difference is smaller than has generally been related to countergradient variation (Conover 1990), and there are confounding elevation and domestication effects. Eagle Lake is 500 m higher than the Mt. Shasta hatchery, potentially shortening the growing season. Also, two strains' extended hatchery propagation makes it hard to distinguish the relative contributions of their genetic heritages and hatchery-based selection.

Table 3. Effects of acclimation temperature on Eagle Lake and Mt. Shasta rainbow trout mean ($\pm$ SEM) critical thermal maxima (CTM). Statistical differences between treatment means are indicated by * between strains at the same temperature and by different numbers for the same strains between temperatures

Year	Temperature (°C)	Strain	Sample size	Weight (g)	CTM (°C)
1996	10	Eagle Lake	19	9.9 $\pm$ 0.77 ^{1*}	27.6 $\pm$ 0.13 ¹
		Mt. Shasta	20	15.7 $\pm$ 1.02 ^{1*}	27.7 $\pm$ 0.11 ¹
	14	Eagle Lake	18	13.2 $\pm$ 1.00 ¹	28.6 $\pm$ 0.14 ^{1,2}
		Mt. Shasta	16	14.5 $\pm$ 1.31 ¹	28.4 $\pm$ 0.17 ^{1,2}
	19	Eagle Lake	18	12.8 $\pm$ 1.16 ¹	29.6 $\pm$ 0.57 ²
		Mt. Shasta	21	12.7 $\pm$ 1.38 ¹	29.3 $\pm$ 0.25 ²
1995	19	Eagle Lake	18	9.2 $\pm$ 1.32 ¹	30.6 $\pm$ 0.22 ¹
		Mt. Shasta	18	12.3 $\pm$ 2.00 ¹	30.0 $\pm$ 0.30 ¹
	22	Eagle Lake	17	7.48 $\pm$ 1.05 ¹	31.0 $\pm$ 0.19 ^{1,2}
		Mt. Shasta	17	11.1 $\pm$ 1.71 ¹	30.7 $\pm$ 0.18 ^{1,2}
	25	Eagle Lake	9	6.1 $\pm$ 1.14 ¹	32.0 $\pm$ 0.19 ²
		Mt. Shasta	9	5.1 $\pm$ 0.62 ¹	31.5 $\pm$ 0.09 ²

Table 4. Temperature effects on Eagle Lake and Mt. Shasta strain rainbow trout mean ($\pm$ SEM) total length (mm), weight (g), and critical swimming velocities (U_{crit}). The same superscripted numbers denote statistical similarity within strains, and the same letters denote statistical similarity between strains

Temperature (°C)	Strain	Sample size	Total Length (mm)	Weight (g)	U_{crit} (m s ⁻¹)	U_{crit} (BL s ⁻¹)
10	Eagle Lake	14	102 $\pm$ 3.9 ^{1a}	11.3 $\pm$ 1.23 ^{1a}	0.46 $\pm$ 0.01 ^{1a}	5.20 $\pm$ 0.10 ^{1a}
	Mt. Shasta	16	120 $\pm$ 2.5 ^{1b}	16.4 $\pm$ 1.49 ^{1b}	0.54 $\pm$ 0.02 ^{1b}	5.27 $\pm$ 0.13 ^{1a}
14	Eagle Lake	16	107 $\pm$ 3.5 ^{1a}	12.2 $\pm$ 1.07 ^{1a}	0.52 $\pm$ 0.02 ^{2a}	5.72 $\pm$ 0.20 ^{1a}
	Mt. Shasta	15	114 $\pm$ 4.0 ^{1a}	15.2 $\pm$ 1.59 ^{1a}	0.53 $\pm$ 0.03 ^{1a}	5.47 $\pm$ 0.31 ^{1a}
19	Eagle Lake	16	113 $\pm$ 5.0 ^{1a}	18.8 $\pm$ 3.17 ^{1a}	0.55 $\pm$ 0.03 ^{2a}	5.66 $\pm$ 0.26 ^{1a}
	Mt. Shasta	16	109 $\pm$ 6.1 ^{1a}	14.3 $\pm$ 2.24 ^{1a}	0.49 $\pm$ 0.04 ^{1a}	5.24 $\pm$ 0.38 ^{1a}

Both strains showed a thermal acclimation effect because critical thermal maxima increased as acclimation temperature increased, as found for other salmonids (Lee and Rinne 1980; Elliott 1991). This effect confers a greater resistance to acute thermal stresses. Because thermal resistance in fish is closely correlated with exposure time (Elliott and Elliott 1995), fish with higher critical thermal maxima can tolerate longer exposures to sub-lethal temperatures, giving them a better chance of escaping to thermal refuges (Matthews et al. 1994) or surviving diel extremes.

We observed no differences between Eagle Lake and Mt. Shasta trouts' thermal tolerances. Critical thermal maxima for both strains appeared to be lower

than those reported for other rainbow trout acclimated to low (10–11 °C), or medium (14–15 °C) temperatures, but were similar to those reported for other salmonids acclimated to higher (19–20 °C) temperatures (Table 5). With the possible exception of lake trout (*Salvelinus namaycush*) (Grande and Andersen 1991), Arctic charr (*S. alpinus*) (Lyytikäinen et al. 1997) and other salmonids restricted to high latitudes, salmonids appear to have very similar thermal tolerances, irrespective of origin (Table 5).

Juvenile Eagle Lake and Mt. Shasta trout critical swimming velocities appeared almost temperature-independent over the 10–19 °C range. The difference between the two strains' critical swimming velocities at 10 °C was due to the larger Mt. Shasta trout;

Table 5. Comparison of mean ($\pm$ SEM) critical thermal maxima (loss of equilibrium endpoint) for various salmonid species. Data marked with (*) are from Lee and Rinne (1980), (†) are from Strange et al. (1993), and (ff) are from Elliott and Elliott (1995). Golden trout data are from Myrick and Cech (submitted), and all remaining data are from this study

Species	Acclimation temperate		
	10–11 °C	14–15 °C	19–20 °C
Brook trout (<i>Salvelinus fontinalis</i>)	28.7 $\pm$ 0.67*	–	29.8 $\pm$ 0.35*
Brown trout (<i>Salmo trutta</i>)	29.0 $\pm$ 0.41*	30.0 $\pm$ 2.0 ^{ff}	29.4 $\pm$ 0.58* 30.0 $\pm$ 1.44 ^{ff}
Atlantic salmon (<i>S. solar</i>)	–	32.9 $\pm$ 2.00 ^{ff}	32.8 $\pm$ 2.02 ^{ff}
Rainbow trout (<i>Oncorhynchus mykiss</i>)	28.5 $\pm$ 0.28*	29.4 $\pm$ 0.08 [†]	29.4 $\pm$ 0.19*
Apache trout (<i>O. apache</i>)	28.5 $\pm$ 0.35*	–	29.4 $\pm$ 0.21*
Gila trout (<i>O. gilae</i>)	28.3 $\pm$ 0.22*	–	29.6 $\pm$ 0.31*
Eagle Lake rainbow trout (<i>O. m. aquilarum</i>)	27.6 $\pm$ 0.13	28.6 $\pm$ 0.14	30.6 $\pm$ 0.19 29.6 $\pm$ 0.57
Mt. Shasta rainbow trout	27.7 $\pm$ 0.11	28.4 $\pm$ 0.17	30.0 $\pm$ 0.19 29.3 $\pm$ 0.25
Little Kern River golden trout (<i>O. m. gilberti</i>)	27.7 $\pm$ 0.12	28.3 $\pm$ 0.30	29.6 $\pm$ 0.96 30.3 $\pm$ 0.08

because length-specific critical swimming velocities were statistically indistinguishable. Both strains' low levels of individual variability were more reminiscent of those observed by Plaut and Gordon (1994) for cloned, compared with wild-type, zebrafish (*Brachydanio rerio*). The relatively low variability may have been the result of reduced genetic diversity or a uniform-rearing environment. Fish of both strains had higher critical swimming velocities than juvenile Nimbus-strain steelhead (Myrick and Cech, in review) and rainbow trout used by Alsop and Wood (1997) (Table 6). The length-specific critical swimming velocities of both Eagle Lake and Mt. Shasta trout were similar to those of Shelton strain coastal cutthroat trout (Hawkins and Quinn 1996) and Little Kern River golden trout (Myrick and Cech, submitted), but were substantially slower than those reported for Aberdeen strain steelhead and coastal cutthroat trout (Table 6) by Hawkins and Quinn (1996).

The short (10-min) interval between stepwise velocity increases may have resulted in artificially high critical swimming velocity values (Hammer 1995), although Beamish (1980) reported no differences for

Arctic charr swum at intervals of 5, 10, and 75 min. Variation among studies may, however, be introduced by differences in fish size, conditioning, experimental protocol, and apparatus design (Hammer 1995).

In conclusion, our study demonstrated that the Eagle Lake and Mt. Shasta strains' physiological responses to temperature were quite similar. Both strains showed direct thermal-dependence regarding their acute thermal tolerances and thermal-independence of critical swimming velocities. Also, their similar food consumption rates, growth rates (except at the highest temperature), oxygen consumption rates, critical swimming velocities, thermal tolerances, and gross conversion efficiencies argue for rejecting our hypothesis concerning significant differences between the strains' physiological performances. However, their demonstrated differences in growth at 22 to 25 °C argue for consideration of strain differences when protecting habitats of unique strains such as Eagle Lake rainbow trout.

Table 6. Comparison of mean length-specific critical swimming velocities for various salmonid species. Data marked with (*) are fork lengths

Species	Temperature (°)	Total length (mm)	Critical swimming velocity (BL s ⁻¹)	Source
Coastal cutthroat trout (Shelton strain)	10	90	5.58	Hawkins and Quinn (1996)
Coastal cutthroat trout (Aberdeen strain)	10	89	6.69	Hawkins and Quinn (1996)
Eagle Lake rainbow trout	10	102	5.2	This study
Little Kern River golden trout	10	85	6	Myrick and Cech (submitted)
Mt. Shasta rainbow trout	10	120	5.27	This study
Rainbow trout	10	287	3.15	Schneider and Connors (1982)
Steelhead (Aberdeen strain)	10	100	7.69	Hawkins and Quinn (1996)
Steelhead (Nimbus strain)	11	136	4.34	Myrick and Cech (in review)
Eagle Lake rainbow trout	14	107	5.72	This study
Little Kern River golden trout	14	89	5.97	Myrick and Cech (submitted)
Mt. Shasta rainbow trout	14	114	5.47	This study
Rainbow trout	15	287	3.37	Schneider and Connors (1982)
Rainbow trout	15	308	1.76	Duthie (1987)
Rainbow trout	15	90–120	3.85	Alsop and Wood (1997)
Steelhead (Nimbus strain)	15	144	4.96	Myrick and Cech (in review)
Eagle Lake rainbow trout	19	113	5.66	This study
Little Kern River golden trout	19	96	4.58	Myrick and Cech (submitted)
Mt. Shasta rainbow trout	19	109	5.24	This study
Steelhead (Nimbus strain)	19	164	4.79	Myrick and Cech (in review)
Rainbow trout	20	287	3.21	Schneider and Connors (1982)

Acknowledgements

This project was funded through a Sport Fishing Institute Fellowship awarded to C.A.M. and PG&E and U.C. Agricultural Experiment Station (3455-H) grants awarded to J.J.C. We thank the California Department of Fish & Game Mt. Shasta State Fish Hatchery for fish used in this study; P. Lutes and W. Bentley for facilities support; G. Dethloff and H. Zhou for statistical assistance; A. Chan, J. Heublein, J. Mesa, T. Mussen, J. Schmidt, M. Sheely, and J. Watters for technical assistance, and; P. Moyle, D. Erman, and S. Railsback for useful discussion and comments on earlier drafts of the manuscript.

References

- Alsop, D.H. and Wood, C.M. 1997. The interactive effects of feeding and exercise on oxygen consumption, swimming performance and protein usage in juvenile rainbow trout (*Oncorhynchus mykiss*). *J. Exp. Biol.* 200: 2337–2346.
- Beamish, F.W.H. 1980. Swimming performance and oxygen consumption of the charrs. In: Charrs: Salmonid Fishes of the Genus *Salvelinus*. pp. 739–748. Edited by E.K. Balon. Dr W. Junk, The Hague.
- Behnke, R.J. 1992. Native trout of western North America. AFS Monograph 6: 275 pp.
- Brett, J.R. 1964. The respiratory metabolism and swimming performance of young sockeye salmon. *J. Fish. Res. Bd. Can.* 21: 1184–1226.
- Brett, J.R. and Groves, T.D.D. 1979. Physiological energetics. In: *Fish Physiology*. Vol. VIII. pp. 279–352. Edited by W.S. Hoar, D.J. Randall and J.R. Brett. Academic Press, New York.
- Brett, J.R., Shelbourn, J.E. and Shoop, C.T. 1969. Growth rate and body composition of fingerling sockeye salmon, *Oncorhynchus nerka*, in relation to temperature and ration size. *J. Fish. Res. Bd. Can.* 26: 2363–2394.
- Briggs, C.T. and Post, J.R. 1997. *In situ* activity metabolism of rainbow trout (*Oncorhynchus mykiss*): estimates obtained from telemetry of axial muscle electromyograms. *Can. J. Fish. Aquat. Sci.* 54: 859–866.
- Busack, C.A. and Gall, G.A.E. 1980. Ancestry of artificially propagated California rainbow trout strains. *Calif. Fish Game* 66: 17–24.
- Cech, J.J., Jr. 1990. Respirometry. In: *Methods for Fish Biology*. pp. 335–362. Edited by C.B. Schreck and P.B. Moyle. American Fisheries Society, Bethesda.
- Cech, J.J., Jr., Mitchell, S.J., Castleberry, D.T. and McEnroe, M. 1990. Distribution of California stream fishes: influence of environmental temperature and hypoxia. *Env. Biol. Fish.* 29: 95–105.
- Cherry, D.S., Dickson, K.L. and Cairns Jr., J. 1977. Preferred, avoided and lethal temperatures of fish during rising temperature conditions. *J. Fish. Res. Bd. Can.* 34: 239–246.
- Conover, D.O. 1990. The relation between capacity for growth and length of growing season: Evidence for and implications of countergradient variation. *Trans. Am. Fish. Soc.* 19: 416–430.
- Conover, D.O., Brown, J.J. and Ehtisham, A. 1997. Countergradient variation in growth of young striped bass (*Morone saxatilis*) from different latitudes. *Can. J. Fish. Aquat. Sci.* 54: 2401–2409.
- Dockray, J.J., Morgan, I.J., Reid, S.D. and Wood, C.M. 1998. Responses of juvenile rainbow trout, under food limitation, to chronic low pH and elevated summer temperatures, alone and in combination. *J. Fish Biol.* 52: 62–82.
- Dockray, J.J., Reid, S.D. and Wood, C.M. 1996. Effects of elevated summer temperatures and reduced pH on metabolism and growth of juvenile rainbow trout (*Oncorhynchus mykiss*) on unlimited ration. *Can. J. Fish. Aquat. Sci.* 53: 2752–2763.
- Elliott, J.M. 1991. Tolerance and resistance to thermal stress in juvenile Atlantic salmon, *Salmo salar*. *Freshw. Biol.* 25: 61–70.
- Elliott, J.M. and Elliott, J.A. 1995. The effect of the rate of temperature increase on the critical thermal maximum for parr of Atlantic salmon and brown trout. *J. Fish Biol.* 47: 917–919.
- Grande, M. and Andersen, S. 1991. Critical thermal maxima for young salmonids. *J. Fresh. Ecol.* 6: 275–279.
- Hammer, C. 1995. Fatigue and exercise tests with fish. *Comp. Biochem. Physiol.* 112A: 1–20.
- Hanson, P., Johnson, J., Kitchell, J. and Schindler, D.E. 1997. Fish Bioenergetics. University of Wisconsin Sea Grant Institute, Madison, Wisconsin.
- Hawkins, D. K. and Quinn, T. P. 1996. Critical swimming velocity and associated morphology of juvenile coastal cutthroat trout (*Oncorhynchus clarki clarki*), steelhead trout (*Oncorhynchus mykiss*), and their hybrids. *Can. J. Fish. Aquat. Sci.* 53: 1487–1496.
- Hokanson, K.E.F., Kellner, C.F. and Thorslund, T.W. 1977. Effects of constant temperatures and diel temperature fluctuations on specific growth and mortality rates and yield of juvenile rainbow trout, *Salmo gairdneri*. *J. Fish. Res. Bd. Can.* 34: 639–648.
- Jobling, M. 1994. Fish Bioenergetics. Chapman and Hall, London.
- Jobling, M. 1997. Temperature and growth: modulation of growth rate via temperature change. In: *Global Warming: Implications for Freshwater and Marine fish*. pp. 225–254. Edited by C. M. Wood and D. G. McDonald. Cambridge University Press, Cambridge.
- Lee, R.M. and Rinne, J.N. 1980. Critical thermal maxima of five trout species in the southwestern United States. *Trans. Am. Fish. Soc.* 109: 632–635.
- Lyttikäinen, T., Koskela, J. and Rissanen, I. 1997. Thermal resistance and upper lethal temperatures of underyearling Lake Inari Arctic charr. *J. Fish Biol.* 51: 515–525.
- Matthews, K.R., Berg, N.H., Azuma, D.L. and Lambert, T.R. 1994. Cool water formation and trout habitat use in a deep pool in the Sierra Nevada, California. *Trans. Am. Fish. Soc.* 123: 549–564.
- McEwan, D. and Jackson, T.A. 1996. Steelhead Restoration and Management Plan for California. California Department of Fish and Game, Sacramento, California.
- Moyle, P.B. 1976. Inland Fishes of California. University of California Press, Berkeley, California.
- Moyle, P.B. and Nichols, R.D. 1974. Decline of the native fish fauna of the Sierra Nevada foothills, central California. *Am. Midl. Nat.* 92: 72–83.
- Ney, J.J. 1993. Bioenergetics modeling today: growing pains on the cutting edge. *Trans. Am. Fish. Soc.* 122: 736–748.
- Nielsen, J.L. Molecular genetics and stock identification in Pacific salmon (*Oncorhynchus* spp.). Doctoral Dissertation, University of California, Berkeley.
- Plaut, I. and Gordon, M.S. 1994. Swimming metabolism of wild-type and cloned zebrafish *Brachydanio rerio*. *J. Exp. Biol.* 194: 209–223.

- Reid, S.D., Linton, T.K., Dockray, J.J., McDonald, D.G. and Wood, C.M. 1998. Effects of chronic sublethal ammonia and a simulated summer global warming scenario: protein synthesis in juvenile rainbow trout (*Oncorhynchus mykiss*). Can. J. Fish. Aquat. Sci. 55: 1534–1544.
- Strange, R.J., Pertie, R.B. and Cech, J.J., Jr. 1993. Slight stress does not lower critical thermal maximums in hatchery-reared rainbow trout. Folia Zool. 42: 251–256.
- Taylor, E.B. and Foote, C.J. 1991. Critical swimming velocities of juvenile sockeye salmon and kokanee, the anadromous and non-anadromous forms of *Oncorhynchus nerka* (Walbaum). J. Fish Biol. 38: 407–419.
- Taylor, E.B. and McPhail, J.D. 1985. Variation in burst and prolonged swimming performance among British Columbia populations of coho salmon, *Oncorhynchus kisutch*. Can. J. Fish. Aquat. Sci. 42: 2029–2033.
- Thomas, A.E. and Donahoo, M.J. 1977. Differences in swimming performance among strains of rainbow trout (*Salmo gairdneri*). J. Fish. Res. Bd. Can. 34: 304–306.
- Wood, C.M. and McDonald, D.G., Editors. 1997. Global Warming: Implications for Freshwater and Marine Fish. Cambridge University Press, Cambridge.
- Wurtsbaugh, W.A. and Davis, G.E. 1977a. Effects of fish size and ration level on the growth and food conversion efficiency of rainbow trout, *Salmo gairdneri* Richardson. J. Fish Biol. 11: 99–104.
- Wurtsbaugh, W.A. and Davis, G.E. 1977b. Effects of temperature and ration level on the growth and food conversion efficiency of *Salmo gairdneri*, Richardson. J. Fish Biol. 11: 87–98.
- Young, P.S. and Cech, J.J., Jr. 1996. Environmental tolerances and requirements of splittail. Trans. Am. Fish. Soc. 125: 664–678.

Reproduced with permission of the copyright owner. Further reproduction prohibited without permission.